

Q1. Write a Python program to create an age_records DataFrame containing the attributes Person_ID and Age for suitable sample records. Perform Data Discretization on the Age attribute using Equal Width Binning. Display the original and discretized data.

[10 Marks]

Q2. Apply Linear Regression on a YouTube Video dataset to predict video views using relevant video attributes. Visualize the relationship between the predictor and target variable using a scatter plot and regression line.

[20 Marks]

OR

Q2. Create the following dataset in python. Apply the Apriori algorithm to generate frequent itemsets and association rules. Repeat the process with different minimum support values 0.2, and 0.3.

[20 Marks]

TID	Items
1	Video Lecture, Quiz
2	Video Lecture, Assignment
3	Quiz, Discussion Forum
4	Video Lecture, Quiz, Assignment
5	Assignment, Discussion Forum
6	Video Lecture, Certificate
7	Quiz, Certificate
8	Video Lecture, Quiz
9	Assignment, Certificate
10	Video Lecture, Quiz, Certificate

Q3. Viva

[5 Marks]